

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250282609

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

MCDONALD; William C. et al.

MICROELECTROMECHANICAL SYSTEM DEVICE WITH OFFSET MIRROR

Abstract

A microelectromechanical system (MEMS) device includes: a mechanical layer; a mirror; and a mirror via coupling the mechanical layer and the mirror. The mirror is laterally offset from the mechanical layer in a direction.

Inventors: MCDONALD; William C. (Allen, TX), HALL; James (Parker, TX), BARTLETT; Terry A. (Dallas, TX)

Applicant: TEXAS INSTRUMENTS INCORPORATED (Dallas, TX)

Family ID: 96948645

Appl. No.: 18/930259

Filed: October 29, 2024

Related U.S. Application Data

us-provisional-application US 63562720 20240308

Publication Classification

Int. Cl.: B81B3/00 (20060101); G02B26/08 (20060101)

U.S. Cl.:

CPC B81B3/0083 (20130101); G02B26/0833 (20130101); B81B2201/042 (20130101); B81B2203/0118 (20130101); B81B2203/0154 (20130101); B81B2203/0361 (20130101); B81B2203/04 (20130101); B81B2207/053 (20130101); B81B2207/094 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Application No. 63/562,720, filed Mar. 8, 2024, entitled “MICROMIRROR DESIGN WITH OFFSET MIRROR FOR CONTRAST IMPROVEMENT.” which is hereby incorporated herein by reference in its entirety.

BACKGROUND

[0002] Some microelectromechanical system (MEMS) devices are used in a spatial light modulator (SLM), such as a digital micromirror device (DMD) to form an array of pixels. An example MEMS device selectively switches a mirror between an on position and an off position responsive to control data. The contrast of a display or projector with an SLM is negatively affected by off-state reflectivity of a MEMS device.

SUMMARY

[0003] In an example, a microelectromechanical system (MEMS) device includes: a mechanical layer; a mirror; and a mirror via coupling the mechanical layer and the mirror, wherein the mirror is laterally offset from the mechanical layer in a direction.

[0004] In another example, a MEMS device includes: an electrode layer; a mechanical layer; hinge vias coupling the electrode layer and the mechanical layer; a mirror; and a mirror via coupling the mechanical layer and the mirror. The mirror has a first position when the MEMS device is in an on state. The mirror has a second position when the MEMS device is in an off state. The mirror is laterally offset relative to the mechanical layer in a direction.

[0005] In yet another example, a MEMS device includes: an electrode layer; a mechanical layer; hinge vias coupling the electrode layer and the mechanical layer; a mirror; and a mirror via coupling the mechanical layer and the mirror, wherein the mirror is laterally offset relative to the mechanical layer in a direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a block diagram of a system in accordance with various examples.

[0007] FIG. 2A is an exploded view of a MEMS device in accordance with various examples.

[0008] FIG. 2B is a perspective view of the MEMS device of FIG. 2A.

[0009] FIG. 3A is an exploded view of a MEMS device in accordance with various examples.

[0010] FIG. 3B is a perspective view of the MEMS device of FIG. 3A.

[0011] FIG. 4 is a cross-sectional view of the MEMS device of FIG. 3B in an on-state.

[0012] FIG. 5 is a cross-sectional view of the MEMS device of FIG. 3B in an off-state.

[0013] FIG. 6 is a top view of the offset mirror relative to a center position for the MEMS device of FIG. 3B.

[0014] FIG. 7A is an incoming light angle view of some of a pixel array with offset mirrors in an on position.

[0015] FIG. 7B is a projection aperture view of the pixel array of FIG. 7A with offset mirrors in an on position.

[0016] FIG. 7C is an incoming light angle view of the pixel array of FIG. 7A with offset mirrors in an off position.

[0017] FIG. 7D is a projection aperture view of the pixel array of FIG. 7A with offset mirrors in an off position.

[0018] FIG. 8 is a perspective view of a MEMS device in accordance with various examples.

[0019] FIG. 9A is a cross-sectional view of the MEMS device of FIG. 8 in an on-state.

[0020] FIG. **9B** is a cross-sectional view of the MEMS device of FIG. **8** in an off-state.

[0021] FIG. **10** is a top view of the offset mirror of the MEMS device of FIG. **9B** relative to a center position.

[0022] FIG. **11A** is incoming light angle view of some of a pixel array with offset mirrors in an on position.

[0023] FIG. **11B** is a projection aperture view of the pixel array of FIG. **11A** with offset mirrors in an on position.

[0024] FIG. **11C** is an incoming light angle view of the pixel array of FIG. **11A** with offset mirrors in an off position.

[0025] FIG. **11D** is a projection aperture view of the pixel array of FIG. **11A** with offset mirrors in an off position.

[0026] FIG. **12** is a top view of sculpted offset mirrors in accordance with various examples.

[0027] FIG. **13** is a flowchart showing a MEMS device fabrication method in accordance with various examples.

DETAILED DESCRIPTION

[0028] The same reference numbers or other reference designators are used in the drawings to designate the same or similar features. Such features may be the same or similar either by function and/or structure.

[0029] Described herein is a microelectromechanical system (MEMS) device with layers. Example layers include an electrode layer, a mechanical layer, and a mirror, where vias are coupled between components of such layers. Example vias include a mirror via, hinge vias, spring tip vias, and raised electrode vias. An example MEMS device also includes a mirror that is offset in a direction relative to a target component (or components) of the MEMS device. Such mirrors are sometimes referred to herein as “offset mirrors”. In some examples, the amount of offset and the direction are selected to increase coverage (reduce exposure) of the target component relative to a projection aperture while the MEMS device is in an off-state. In some examples, the target component is a reflective component of a mechanical layer, such as a hinge, a raised electrode, and/or a spring tip. Increasing coverage of the target component while the MEMS device is in the off-state relative to the projection aperture, reduces the amount of light that reaches the projection aperture while the MEMS device is in the off-state. The result is a darker off-state for MEMS device relative to the projection aperture, resulting in improved on/off contrast for a MEMS device.

[0030] In some examples, each MEMS device is a pixel. In different examples, such MEMS devices are organized into a pixel array for a spatial light modulator (SLM), such as a digital micromirror device (DMD). During operations, the SLM receives data from a controller to control on/off states of pixels of the pixel array. In the off-state, the offset mirrors reduce reflectivity of respective target components relative to a projection aperture. In some examples, the projection aperture is part of a projector or display that includes the SLM, the controller, and a light source. By using a pixel array with offset mirrors, the contrast of a projector or display is increased (e.g., up to 20% or more) without significant redesign of a MEMS device. For example, a MEMS device layout may be modified to include an offset mirror without changing the layout of mechanical layer components and/or electrode layer components. In other examples, a MEMS device layout may include some modifications to mechanical layer components and/or electrode layer components, where use of an offset mirror reduces the amount of modification needed to achieve a target contrast.

[0031] In some examples, a mirror via center is aligned with an offset mirror center and is not aligned with a mechanical layer center. In such examples, aligning the mirror via center with the offset mirror center and not with the mechanical layer center may involve some changes to the mechanical layer layout (e.g., increasing cost and limiting spacing and/or size reduction options for MEMS device fabrication) while ensuring the offset mirror is balanced on the mirror via and has consistent movement. In other examples, a mirror via center is aligned with a mechanical layer

center and is not aligned with an offset mirror center. In such examples, aligning the mirror via center with the mechanical layer and not with the offset mirror center facilitates mechanical layer layout (e.g., maintaining cost, spacing and/or size reduction options for MEMS device fabrication) while possibly increasing offset mirror imbalance and related issues.

[0032] FIG. 1 is a block diagram of a system **100** in accordance with various examples. In some examples, system **100** is a projector, for example a traditional projector, an augmented reality (AR) display, a virtual reality (VR) display, a smart headlight, a heads-up display (HUD), an automotive ground projector, a light detection and ranging (LIDAR) unit, a lithography unit, a three-dimensional (3D) printer, a spectroscopy display, a 3D display, or another type of projector. The system **100** may also represent some or all of a display such as a DMD display.

[0033] As shown, system **100** includes a controller **102**, a light source **120**, an SLM **128**, and a projection aperture **138**. The controller **102** has a first terminal **104**, a second terminal **106**, a third terminal **108**, and a fourth terminal **109**. The light source **120** has an input **122** and an optical output **124**. The SLM **128** has an input **130**, an optical input **132**, and an optical output **134**. In the example of FIG. 1, the SLM **128** includes a pixel array **136** having pixels with offset mirrors as described herein. In different examples, the SLM **128** may perform spatial modulation of light using mechanical, electro-optical, thermo-optical, and/or magneto-optical control options.

[0034] In the example of FIG. 1, the controller **102** operates to: receive video data at the first terminal **104**; receive configuration data at the second terminal **106**; provide control signals CS1 at the third terminal **108** responsive to the video data and the configuration data; and provide control signal CS2 at the fourth terminal **109** responsive to the video and the configuration data. The light source **120** operates to: receive the control signals CS1 at the input **122**; and generate light **126** at the optical output **124** responsive to the control signal CS1. In some examples, the light source **120** modulates the intensity, color, and/or timing of the light **126** at the optical output **124** responsive to the control signal CS1. The SLM **128** operates to: receive the control signals CS2 at the input **130**; receive the light **126** at the optical input **132**; and provide spatially-modulated light at the optical output **134** responsive to the light **126** and the control signals CS2. In some examples, the control signals CS2 include bit plane (BP) data and control signals to control light modulation options of the SLM **128**. Without limitation, the control signals CS2 may be transferred to the SLM **128** using low-voltage differential signaling (LVDS). The spatially-modulated light from the SLM **128** results in projected video **140**.

[0035] With the pixel array **136** and related offset mirrors, the reflectivity of MEMS device components of the SLM **128** relative to the projection aperture **138** is reduced when the MEMS devices are in an off-state. This reduction in reflectivity results in the projected video **140** having an improved contrast relative to an SLM having a pixel array without offset mirrors.

[0036] FIG. 2A is an exploded view **200** of a MEMS device **202** in accordance with various examples. The MEMS device **202** is an example of a pixel of the pixel array **136** of the SLM **128** in FIG. 1. In some examples, the MEMS device **202** is part of a dual spring tip pixel as in FIGS. 3A to 3E. In other examples, the MEMS device **202** is part of a cantilever hinge pixel as in FIG. 8.

[0037] In the example of FIG. 2A, the MEMS device **202** includes a base **204**, an electrode layer **206**, via(s) **212**, a mechanical layer **208**, via(s) **214**, and an offset mirror **211**. The base **204** includes memory cells (not shown) to control different states of the MEMS device **202** responsive to received data. In some examples, the electrode layer **206** includes first and second electrodes coupled to the base **204**. In some examples, via(s) **212** may include electrode vias, spring tip vias, and/or hinge vias. In some examples, the mechanical layer **208** includes one or more hinges, raised electrodes, spring tips, and/or other components. In some examples, via(s) **214** may include a mirror via. In some examples, the via(s) **212** couple the mechanical layer **208** to the electrode layer **206**. Additionally, or alternatively, the via(s) **214** may couple the mechanical layer **208** to the offset mirror **211**.

[0038] In the example of FIGS. 2A and 2B, the offset mirror **211** is offset relative to a center

position **209**. In some examples, the center position **209** aligns a center point **210** of the center position **209** with a mirror via center (e.g., a center point of one of the via(s) **214**), a layer center (e.g., a center point of the mechanical layer **208** and/or a center point of the electrode layer **206**), and/or a center line **232** of MEMS device **20**. The center line **232** of the MEMS device **202** may be aligned, for example, with a via center, a first layer center (e.g., a center point of the electrode layer **206**) and/or a second layer center (e.g., a center point of the mechanical layer **208**). As another option, the center position **209** is aligned with boundaries **230A** and **230B** of the MEMS device **202**. The boundaries **230A** and **230B** are selected, for example, based on outer boundaries of the electrode layer **206** and/or the mechanical layer **208**. In the example of FIGS. **2A** and **2B**, the offset mirror **211** is offset from the center position **209** in the **Y1** direction to increase coverage of a target component relative to a projection aperture while the MEMS device **202** is in an off-state. In some examples, the target component is a mechanical layer component (e.g., a hinge, a spring tip, and/or a raised electrode).

[0039] FIG. **2B** is a perspective view **250** of the MEMS device **202** of FIG. **1A** in accordance with various examples. In the example of FIG. **2B**, the MEMS device **202** includes the base **204**, the electrode layer **206**, the mechanical layer **208**, and the offset mirror **211** together. In the example of FIG. **2B**, the via(s) **212** couple the electrode layer **206** and the mechanical layer **208**. Also, the via(s) **214** couple the mechanical layer **208** and the offset mirror **211**. With the offset mirror **211** offset in the **Y1** direction, exposure of a target component to the projection aperture **138** is reduced compared to a mirror having the center position **209**.

[0040] Reducing reflections from a target component while the MEMS device **202** is in an off-state improves the contrast between on/off states of the MEMS device **202** relative to the projection aperture **138** (i.e., use of the offset mirror **211** results in a darker off-state of the MEMS device **202** as viewed from the projection aperture **138** compared to use of a mirror having the center position **209**). With a mirror at the center position **209**, the target component may be a mechanical layer component (e.g., a hinge, a spring tip, a raised electrode, etc.) that is reflective and at least partially exposed relative to the projection aperture **138** while the MEMS device **202** is in an off-state. Such exposure of the target component can result in some reflected light being directed towards the projection aperture **138** while the MEMS device **202** is in the off-state. With the offset mirror **211** offset in the **Y1** direction instead of a mirror in the center position **209**, the exposure of the target component relative to the projection aperture while the MEMS device **202** is in the off-state is reduced or eliminated. This reduction in exposure of the target component while the MEMS device **202** results in a darker off-state for the MEMS device, improving overall contrast between the on-state and the off-state of the MEMS device **202**.

[0041] In the example of FIGS. **2A** and **2B**, the MEMS device **202** is oriented based on an **X1** direction and a **Y1** direction. With the orientation represented in FIGS. **2A** and **2B**, the offset mirror **211** is offset towards the **Y1** direction relative to the center position **209**. In other examples, depending on the layout of mechanical layer components, the offset mirror **211** may be offset towards a $-Y1$ direction, the **X1** direction, the $-X1$ direction, or a combination of directions (e.g., an **X1** and **Y1** direction, an **X1** and $-Y1$ direction, a $-X1$ and **Y1** direction, or a $-X1$ and $-Y1$ direction).

[0042] FIGS. **3A** and **3B** are different views **330** and **350** of an example MEMS device **300**. The MEMS device **300** is an example of a dual spring tip pixel of the pixel array **136** of the SLM **128** in FIG. **1**, or the MEMS device **202** in FIGS. **2A** and **2B**. FIG. **3A** is an exploded view **330** of the MEMS device **300**. FIG. **3B** is a perspective view **350** of the MEMS device **300**. In the example of FIG. **3A**, the MEMS device **300** includes a base **301**, a first address electrode **302A**, a second address electrode **302B**, a bias electrode **304**, spring tip vias **314A**, **314B**, **314C**, and **314D**, first electrode vias **310A**, second electrode vias **310B**, hinge vias **306**, a torsion hinge **308**, spring tips **316A** to **316D**, a first raised electrode **312A**, a second raised electrode **312B**, a mirror via **318**, and an offset mirror **311**.

[0043] In the example of FIG. 3A, the base **301** is an example of the base **204** in FIGS. 2A and 2B. The first address electrode **302A**, the second address electrode **302B**, and the bias electrode **304** are example components of an electrode layer **321**. The electrode layer **321** is an example of the electrode layer **206** in FIGS. 2A and 2B. The spring tip vias **314A**, **314B**, **314C**, and **314D**, the first electrode vias **310A**, the second electrode vias **310B**, and the hinge vias **306** are examples of the via(s) **212** in FIGS. 2A and 2B. The torsion hinge **308**, the spring tips **316A** to **316D**, the first raised electrode **312A**, and the second raised electrode **312B** are example components of a mechanical layer **324**. The mechanical layer **324** is an example of the mechanical layer **208** in FIGS. 2A and 2B. The mirror via **318** is an example of the via(s) **214** in FIGS. 2A and 2B. The offset mirror **311** is an example of the offset mirror **211** in FIGS. 2A and 2B.

[0044] In the example of FIGS. 3A and 3B, the offset mirror **311** is offset relative to a center position **309**. In some examples, the center position **309** aligns a center point **313** of the center position **309** with a mirror via center (e.g., a center point of the mirror via **318**), a layer center (e.g., a center point of the mechanical layer **324** and/or a center point of the electrode layer **321**), and/or a center line **332** of MEMS device **300**. The center line **332** of the MEMS device **300** may be aligned, for example, with a via center, a first layer center (e.g., a center point of the electrode layer **321**) and/or a second layer center (e.g., a center point of the mechanical layer **324**). As another option, the center position **309** is aligned with boundaries **330A** and **330B** of the MEMS device **300**. The boundaries **330A** and **330B** are selected, for example, based on outer boundaries of the electrode layer **321** and/or the mechanical layer **324**. In the example of FIGS. 3A and 3B, the offset mirror **311** is offset from the center position **309** in the Y2 direction to increase coverage of a target component relative to a projection aperture while the MEMS device **300** is in an off-state. In some examples, the target component is a mechanical layer component (e.g., the torsion hinge **308**, one or more of the spring tips **316A** to **316D**, the first raised electrode **312A**, and the second raised electrode **312B**).

[0045] In the example of FIG. 3B, the position of the offset mirror **311** is tilted towards the spring tips **316A** and **316B**, which may be an on position for the offset mirror **311**. In an example SLM, the position of the offset mirror **311** switches between different tilted positions (e.g., an on position in which the offset mirror **311** contacts spring tips **316A** and **316B**, and an off position in which the offset mirror **311** contacts the spring tips **316C** and **316C**) responsive to: received data; control voltages applied to the first address electrode **302A**, the second address electrode **302B**, and the bias electrode **304** responsive to received data; and movement of the torsion hinge **308** and offset mirror **311** responsive to application of the control voltages.

[0046] In the example of FIG. 3B, the offset mirror **311** is in a first position, in which the offset mirror **311** contacts the spring tips **316A** and **316B** at contact points **322** responsive to control voltages applied to the first address electrode **302A**, the second address electrode **302B**, and the bias electrode **304**. To change the position of the offset mirror **311** to another position (e.g., the offset mirror **311** may contact the spring tips **316C** and **316D**), updated control voltages are applied to the first address electrode **302A**, the second address electrode **302B**, and/or the bias electrode **304**.

[0047] In the example of FIGS. 3A and 3B, the MEMS device **300** is oriented based on an X2 direction and a Y2 direction. With the orientation represented in FIGS. 3A and 3B, the offset mirror **311** is offset towards the Y2 direction. In other examples, the offset mirror **311** may be offset towards a -Y2 direction, the X2 direction, the -X2 direction, or a combination of directions (e.g., an X2 and Y2 direction, an X2 and -Y2 direction, a -X2 and Y2 direction, or a -X2 and -Y2 direction).

[0048] FIG. 4 is a cross-sectional view **400** of the MEMS device **300** of FIG. 3B in an on-state. In the cross-sectional view **400** of FIG. 4, the base **301**, the first address electrode **302A**, the second address electrode **302B**, the bias electrode **304**, the first raised electrode **312A**, the second raised electrode **312B**, the mirror via **318**, and the offset mirror **311** of the MEMS device **300** are

represented. Also represented in FIG. 4 is a projection aperture **448**, a light source **402**, a light sink **404**, incoming light **460** (e.g., from the light source **402**, outgoing light **462**, and a mirror normal direction **466**. The light source **402** is an example of the light source **120** in FIG. 1. The projection aperture **448** is an example of the projection aperture **138** in FIG. 1. The mirror normal direction **466** is perpendicular to the surface of the offset mirror **311** in the on position.

[0049] When the MEMS device **300** is in the on-state as in the example of FIG. 4, the offset mirror **311** is in an on position. With the offset mirror **311** in the on position, the outgoing light **462**, resulting from the incoming light **460** reflecting from the surface of the offset mirror **311**, is directed towards the projection aperture **448** and not the light sink **404**. The offset mirror **311** in the on position does not significantly alter the amount of the outgoing light **462** directed to the projection aperture **448** relative to a mirror having a center position (see e.g., the center position **209** in FIG. 2, or the center position **309** in FIG. 3). However, in an off position, the offset mirror **311** reduces the amount of stray light (e.g., light remaining after the on-state due to reflections or other design issues) directed to the projection aperture **448**.

[0050] FIG. 5 is a cross-sectional view **500** of the MEMS device **300** of FIG. 3B in an off-state. In the cross-sectional view **500** of FIG. 5, the base **301**, the first address electrode **302A**, the second address electrode **302B**, the bias electrode **304**, the first raised electrode **312A**, the second raised electrode **312B**, the mirror via **318**, and the offset mirror **311** of the MEMS device **300** are represented. Also represented in FIG. 5 is the projection aperture **448**, the light source **402**, the light sink **404**, first stray light **540A** (e.g., stray light remaining the on-state of FIG. 4), first outgoing light **542A**, second stray light **540B** (e.g., stray light remaining the on-state of FIG. 4), second outgoing light **542B**, and a mirror normal direction **546**. The mirror normal direction **546** is perpendicular to the surface of the offset mirror **311** in the off position.

[0051] When the MEMS device **300** is in the off-state, the offset mirror **311** is in the off position. With the offset mirror **311** in the off position, the first outgoing light **542A**, resulting from the first stray light **540A** reflecting from the surface of the offset mirror **311**, is directed away from the projection aperture **448** to the light sink **404**. However, the second outgoing light **542B**, resulting from the second stray light **540B** reflecting from surfaces below the offset mirror **311** (e.g., the first raised electrode **312A**, other mechanical layer components, and/or electrode layer components), is directed towards the projection aperture **448**. The result of the second outgoing light **542B** being directed to the projection aperture **448** while the MEMS device **300** is in the off-state is that the contrast (the difference between on-state lighting versus off-state lighting of the MEMS device **300**) is reduced. With the offset mirror **311**, the amount of second outgoing light **542B** directed to the projection aperture **448** is reduced compared to when a non-offset mirror is used. This is because the offset mirror **311** reduces exposure of a reflective target component (e.g., the torsion hinge **308**, the spring tips **316A** to **316D**, the first raised electrode **312A**, and/or the second raised electrode **312B**) to the projection aperture **448**. In the example of FIG. 5, the second outgoing light **542B** is represented as being reflected from the first raised electrode **312A**. In other examples, outgoing light directed to the projection aperture **448** is due to stray light being reflected from a spring tip (e.g., one or more of the spring tips **316A** to **316D**).

[0052] In the example of FIG. 5, the offset mirror **311** is offset in the Y2 direction to cover more of a target mechanical layer component (e.g., the first raised electrode **312A**, the second raised electrode **312B**, a torsion hinge **308** as in FIGS. 3A and 3B, and/or springs tips **316A**, **316B**, **316C**, and/or **316D**), a target electrode layer component (e.g., the first address electrode **302A**, the second address electrode **302B**, and/or the bias electrode **304**), and/or a via (e.g., the mirror via **318**, the hinge vias **306**, the spring tip vias **314A**, **314B**, **314C**, and/or **314D**, the first electrode vias **310A** and/or second electrode vias **310B**).

[0053] FIG. 6 is a top view **600** of the offset mirror **311** of the MEMS device **300** of FIG. 3B relative to the center position **309**. In the example of FIG. 6, the center position **309** has the center point **313** and sides A, B, C, and D. The offset mirror **311** also has a center point **612** and sides A,

B, C, and D. In some examples, the center position **309** aligns the center point **313** with a mirror via center (e.g., a center point of the mirror via **318** in FIGS. **3A** and **3B**) and/or a layer center (e.g., a center point of the mechanical layer **324** in FIG. **3A** and/or a center point of the electrode layer **321** in FIG. **3A**). As another option, the center position **309** is aligned with boundaries **630A** and **630B** in FIG. **6**. The boundaries **630A** and **630B** are selected, for example, based on outer boundaries of the electrode layer **321** of the MEMS device **300** and/or of the mechanical layer **324** of the MEMS device **300**. In the example of FIG. **6**, the offset mirror **311** is offset from the center position **309** or the center point **313** of the center position **309** in the Y2 direction by a spacing (OFFSET1). OFFSET1 is also the difference between the center point **313** of the center position **309** and the center point **612** of the offset mirror **311**. With OFFSET1, side A of center position **309** is offset from side A of the offset mirror **311** by spacing **a1**, side B of center position **309** is offset from side B of the offset mirror **311** by spacing **a1**, side C of center position **309** is offset from side C of the offset mirror **311** by spacing **a1**, and side D of center position **309** is offset from side D of the offset mirror **311** by spacing **a1**. Assuming a square shape for the offset mirror **311** and a perpendicular intersection, the distance from the center point **313** of the center position **309** to side A of the offset mirror **311** is $L/2 - a1$, where L is the length of each side of the offset mirror **311**. The distance from the center point **313** of the center position **309** to a perpendicular intersection of side B of the offset mirror **311** is also $L/2 - a1$. The distance from the center point **313** of the center position **309** to a perpendicular intersection of side C of the offset mirror **311** is also $L/2 + a1$. The distance from the center point **313** of the center position **309** to a perpendicular intersection of side D of the offset mirror **311** is also $L/2 + a1$. In the example of FIG. **6**, $OFFSET1 = \sqrt{2} * a1$. . .

[0054] With OFFSET1 in the Y2 direction and the related spacings **a1**, the offset mirror **311** improves coverage of a target mechanical layer component (e.g., the first raised electrode **312A**, the second raised electrode **312B**, a torsion hinge **308** as in FIGS. **3A** and **3B**, and/or springs tips **316A**, **316B**, **316C**, and/or **316D**), a target electrode layer component (e.g., the first address electrode **302A**, the second address electrode **302B**, and/or the bias electrode **304**), and/or a via (e.g., the mirror via **318**, the hinge vias **306**, the spring tip vias **314A**, **314B**, **314C**, and/or **314D**, first electrode vias **310A** and/or second electrode vias **310B**) relative to a projection aperture. In some examples, OFFSET1 in the Y2 direction is approximately 0.212 μm and the spacing **a1** is approximately 0.15 μm for a 9.0 μm pixel size. In this first example, OFFSET1 is approximately 2.35% of the pixel size. In another example, OFFSET1 in the Y2 direction is approximately 0.106 micrometers and the spacing **a1** is approximately 0.075 micrometers for a 9.0 μm pixel size. In this second example, OFFSET1 is approximately 1.18% of the pixel size. In different examples, the offset, the spacing **a1**, and/or the pixel size may vary. Example pixel sizes may range from 16 μm pixels down to 2.7 μm pixels or smaller.

[0055] FIGS. **7A** to **7D** are different views **700**, **710**, **720**, and **730** of a pixel array **702** based on MEMS devices **300A** to **300E** with respective offset mirrors **311A** to **311E**. Each of the MEMS devices **300A** to **300E** is an example of the MEMS device **300** in FIGS. **3A**, **3B**, **4**, and **5**. Each of the respective offset mirrors **311A** to **311E** is an example of the offset mirror **311** in FIGS. **3A**, **3B**, **4**, **5**, and **6**. In FIGS. **7A** to **7D**, the offset mirrors **311A** to **311E** are offset in the Y2 direction relative to a center position (e.g., the center position **309** in FIG. **6**).

[0056] FIG. **7A** is an incoming light angle view **700** of part of the pixel array **702** with the offset mirrors **311A** to **311E** in an on position (the mirror land direction or tilt for the on position is in the Y2 or down direction in FIG. **7A**). In the example of FIG. **7A**, the offset mirrors **311A** to **311E**, and electrode layer component portions **704A** to **704E** of the respective MEMS devices **300A** to **300E** are represented. In FIGS. **7A** to **7D**, the offset mirrors **311A** to **311E** are offset relative to other MEMS device layers including respective electrode layers (e.g., each respective electrode layer represented by the electrode layer **321** in FIG. **3A**) and respective mechanical layers (e.g., each respective mechanical layer represented by the mechanical layer **324** in FIGS. **3A**). Each of the electrode layer component portions **704A** to **704E** may be part of a first address electrode (e.g., the

first address electrode **302A** in FIGS. **3A**, **3B**, **4**, and **5**), part of a second address electrode (e.g., the second address electrode **302B** in FIGS. **3A**, **3B**, **4**, and **5**), or part of a bias electrode (e.g., the bias electrode **304** in FIGS. **3A**, **3B**, **4**, and **5**).

[0057] FIG. **7B** is a projection aperture view **710** of the pixel array **702** with the offset mirrors **311A** to **311E** in an on position (the mirror land direction or tilt for the on position is in the Y2 or down direction in FIG. **7B**). In the example of FIG. **7B**, the offset mirrors **311A** to **311E**, the electrode layer component portions **704A** to **704E**, and mechanical layer component portions **708A** to **708E** of the respective MEMS devices **300A** to **300E** are represented. Each of the mechanical layer component portions **708A** to **708E** may be part of a torsion hinge (e.g., the torsion hinge **308** in FIGS. **3A**, **3B**, **4**, and **5**), part of a spring tip (e.g., the spring tips **316A** to **316D** in FIGS. **3A** and **3B**), part of a first raised electrode (e.g., the first raised electrode **312A** in FIGS. **3A**, **3B**, **4**, and **5**), or part of a second raised electrode (e.g., the second raised electrode **312B** in FIGS. **3A**, **3B**, **4**, and **5**).

[0058] FIG. **7C** is an incoming light angle view **720** of the pixel array **702** with the offset mirrors **311A** to **311E** in an off position (the mirror land direction or tilt for the off position is in the -Y2 or up direction in FIG. **7C**). In the example of FIG. **7C**, the offset mirrors **311A** to **311E**, the electrode layer component portions **704A** to **704E**, and the mechanical layer component portions **708A** to **708E** of the respective MEMS devices **300A** to **300E** are represented.

[0059] FIG. **7D** is a projection aperture view **730** of the pixel array **702** with the offset mirrors **311A** to **311E** in the off position (the mirror land direction or tilt for the off position is in the -Y2 or up direction in FIG. **7D**). In the example of FIG. **7C**, the offset mirrors **311A** to **311E**, the electrode layer component portions **704A** to **704E**, and the mechanical layer component portions **708A** to **708E** of the respective MEMS devices **300A** to **300E** are represented.

[0060] In the incoming light angle view **700** of FIG. **7A**, with the offset mirrors **311A** to **311E** in the on position, most MEMS device components below the offset mirrors **311A** to **311E** are not visible except some electrode layer component portions such as the electrode layer component portions **704A** to **704E**. In the projection aperture view **710** of FIG. **7B**, with the offset mirrors **311A** to **311E** in the on position, most MEMS device components below the offset mirrors **311A** to **311E** are not visible except some electrode layer component portions such as the electrode layer component portions **704A** to **704E** and some mechanical layer component portions such as the mechanical layer component portions **708A** to **708E**. In the incoming light angle view **720** of FIG. **7C**, with the offset mirrors **311A** to **311E** in the off position, most MEMS device components below the offset mirrors **311A** to **311E** are not visible except some electrode layer component portions such as the electrode layer component portions **704A** to **704E** and some mechanical layer component portions such as the mechanical layer component portions **708A** to **708E**. Although more of the electrode layer component portions **704A** to **704E** and more of the mechanical layer component portions **708A** to **708E** are visible in the incoming light angle view **720** of FIG. **7C**, the incoming light view **720** does not significantly affect the amount of stray light directed to a projection aperture. Exposure of electrode layer component portions **704A** to **704E** and/or the mechanical layer component portions **708A** to **708E** relative to a projection aperture is more relevant than exposure from an incoming light view. In the projection aperture view **730** of FIG. **7D**, with the offset mirrors **311A** to **311E** in the off position, most MEMS device components below the offset mirrors **311A** to **311E** are not visible except some electrode layer component portions such as the electrode layer component portions **704A** to **704E** and some mechanical layer component portions such as the mechanical layer component portions **708A** to **708E**. Compared to mirrors aligned with center positions (e.g., the center position **209** in FIGS. **2A** and **2B**, or the center position **309** in FIGS. **3A**, **3B**, and **6**), the offset mirrors **311A** to **311E** cover more electrode layer component portions (e.g., more of the electrode layer component portions **704A** to **704E**) and/or mechanical layer component portions (e.g., more of the mechanical layer component portions **708A** to **708E**) relative to a projection aperture, which improves contrast.

[0061] Compared to mirrors in a centered position, the offset mirrors **311A** to **311E** in FIGS. 7A to 7D cover more mechanical layer component portions such as the mechanical layer component portions **708A** to **708E**. Compared to mirrors in a centered position, the offset mirrors **311A** to **311E** in FIGS. 7A to 7D may also cover more electrode layer component portions such as the electrode layer component portions **704A** to **704E**. To improve contrast of a display or projector, increased coverage of mechanical layer components for the projection aperture view **730** of FIG. 7D has highest priority. This is because scattered light due to visibility of mechanical layer components in the projection aperture view **730** reaches the aperture (e.g., the projection aperture **138**). While full coverage of mechanical layer components for the projection aperture view **730** is optimal, any coverage improvement of mechanical layer components (relative to centered mirrors) can provide a contrast improvement.

[0062] In some examples, electrode layer components are treated with a non-reflective coating, while mechanical layer components (e.g., the torsion hinge **308**, the spring tips **316A** to **316D**, the first raised electrode **312A**, and/or the second raised electrode **312B**) are untreated. In some examples, treating mechanical layer components with a non-reflective coating is avoided to ensure consistent movement of the mechanical layer components. In the described examples, the reflectivity of untreated mechanical layer components may result in some extraneous reflected light being directed to a projection aperture. In the described examples, the amount of extraneous reflected light due to mechanical layer components is reduced, at least in a projection aperture view such as the projection aperture view **730**, using offset mirrors (e.g., the offset mirrors **311A** to **311E**).

[0063] FIG. 8 is a perspective view of a MEMS device **800** in accordance with various examples. The MEMS device **800** is an example of a cantilever hinge pixel of the pixel array **136** of the SLM **128** in FIG. 1, or the MEMS device **202** in FIGS. 2A and 2B. In the example of FIG. 8, the MEMS device **800** includes a base **801**, a first address electrode **802A**, a second address electrode **802B**, a bias electrode **804**, spring tip vias **814A**, **814B**, and **814C**, a first electrode via **810A**, a second electrode via **810B**, hinge vias **806**, a cantilever hinge **808**, spring tips **816A** to **816C**, a first raised electrode **812A**, a second raised electrode **812B**, a mirror via **818**, and an offset mirror **811**.

[0064] In the example of FIG. 8, the base **801** is an example of the base **204** in FIGS. 2A and 2B. The first address electrode **802A**, the second address electrode **802B**, and the bias electrode **804** are example components of an electrode layer such as the electrode layer **206** in FIGS. 2A and 2B. The spring tip vias **814A**, **814B**, and **814C**, the first electrode via **810A**, the second electrode via **810B**, and the hinge vias **806** are example vias such as the via(s) **212** in FIGS. 2A and 2B. The cantilever hinge **808**, the spring tips **816A** to **816C**, the first raised electrode **812A**, and the second raised electrode **812B** are example components of a mechanical layer such as the mechanical layer **208** in FIGS. 2A and 2B. The mirror via **818** is an example of via(s) **214** in FIGS. 2A and 2B. The offset mirror **811** is an example of the offset mirror **211** in FIGS. 2A and 2B.

[0065] In the example of FIG. 8, the position of the offset mirror **811** is tilted towards the spring tips **816B** and **816C**, which may be an on position for the offset mirror **811**. In an example SLM, the position of the offset mirror **811** switches between different tilted positions (e.g., an on position in which the offset mirror **811** contacts spring tips **816B** and **816C**, and an off position in which the offset mirror **811** contacts the spring tips **816A** and **816C**) responsive to: received data; control voltages applied to the first address electrode **802A**, the second address electrode **802B**, and the bias electrode **804** responsive to received data; and movement of the cantilever hinge **808** and offset mirror **811** responsive to application of the control voltages.

[0066] In the example of FIG. 8, the offset mirror **811** is in a first position, in which the offset mirror **811** contacts the spring tips **816B** and **816C** at contact points **822** responsive to control voltages applied to the first address electrode **802A**, the second address electrode **802B**, and the bias electrode **804**. To change the position of the offset mirror **811** to another position (e.g., the offset mirror **811** may contact the spring tips **816A** and **816C**), updated control voltages are applied

to the first address electrode **802A**, the second address electrode **802B**, and/or the bias electrode **804**. In the example of FIGS. **8**, the MEMS device **800** is oriented based on an **X3** direction and a **Y3** direction. Also, the offset mirror **811** is offset relative to a center position **809**. In some examples, the center position **809** aligns a center point **813** of the center position **809** with a mirror via center (e.g., a center point of the mirror via **818**), a layer center (e.g., a center point of the mechanical layer of the MEMS device **800** and/or a center point of the electrode layer of the MEMS device **800**), and/or a center line **332** of MEMS device **300**. As another option, the center position **309** is aligned with boundaries of the MEMS device **800**. Such boundaries may be selected, for example, based on outer boundaries of the electrode layer and/or the mechanical layer of the MEMS device **800**. In the example of FIG. **8**, the offset mirror **811** is offset from the center position **809** in the **X3** and **Y3** directions to increase coverage of a target component relative to a projection aperture while the MEMS device **800** is in an off-state. In some examples, the target component is a mechanical layer component (e.g., the cantilever hinge **808**, one or more of the spring tips **816A** to **816C**, the first raised electrode **812A**, and the second raised electrode **812B**). With the orientation represented in FIG. **8**, the offset mirror **811** is offset towards the $-X3$ and $-Y3$ directions. In other examples, the offset mirror **311** may be offset towards a **X3** direction, a $-X3$ direction, a **Y3** direction, a $-Y3$ direction, or a combination of directions (e.g., an **X3** and **Y3** direction, an **X3** and $-Y3$ direction, or a $-X3$ and **Y3** direction).

[0067] FIG. **9A** is a cross-sectional view **900** of the MEMS device **800** of FIG. **8** in an on-state. In the cross-sectional view **900** of FIG. **9A**, the base **801**, the bias electrode **804**, the spring tips **816A** and **816B**, the mirror via **818**, and the offset mirror **811** are represented. Also represented in FIG. **9A** is a projection aperture **948**, a light source **902**, a light sink **904**, incoming light **930** (e.g., from the light source **902**), outgoing light **932**, and a first mirror normal direction **966A**. The light source **902** is an example of the light source **120** in FIG. **1**. The projection aperture **948** is an example of the projection aperture **138** in FIG. **1**. The first mirror normal direction **966A** is perpendicular to the surface of the offset mirror **811** in the on position. When the MEMS device **800** is in the on-state as in the example of FIG. **9A**, the offset mirror **811** is in an on position.

[0068] With the offset mirror in the on position, the outgoing light **932**, resulting from the incoming light **930** reflecting from the surface of the offset mirror **811**, is directed towards the projection aperture **948** and not the light sink **904**. The offset mirror **811** in the on position does not significantly alter the amount of the outgoing light **932** directed to the projection aperture **948** relative to a mirror having a center position (see e.g., the center position **809** in FIG. **8**). However, in an off position, the offset mirror **811** reduces the amount of stray light (e.g., light remaining after the on-state due to reflections or other design issues) directed to the projection aperture **948**.

[0069] FIG. **9B** is a cross-sectional view **950** of the MEMS device **800** of FIG. **8** in an off-state. In the cross-sectional view **950** of FIG. **9B**, the base **801**, the bias electrode **804**, the spring tips **816A** and **816B**, the mirror via **818**, and the offset mirror **811** of the MEMS device **800** are represented. Also represented in FIG. **9B** is the projection aperture **948**, the light source **902**, the light sink **904**, first stray light **940A** (e.g., stray light remaining the on-state of FIG. **9A**), first outgoing light **942A**, second stray light **940B** (e.g., stray light remaining the on-state of FIG. **9A**), second outgoing light **942B**, and a second mirror normal direction **966B**. The second mirror normal direction **966B** is perpendicular to the surface of the offset mirror **811** in the off position. When the MEMS device **800** is in the off-state, the offset mirror **811** is in the off position. With the offset mirror **811** in the off position, the first outgoing light **942A**, resulting from the first stray light **940A** reflecting from the surface of the offset mirror **811**, is directed away from the projection aperture **948** to the light sink **904**. However, the second outgoing light **942B**, resulting from the second stray light **940B** reflecting from surfaces below the offset mirror **811** (e.g., the spring tips **816A**, other mechanical layer components, and/or electrode layer components), is directed towards the projection aperture **948**. The result of the second outgoing light **942B** being directed to the projection aperture **948** while the MEMS device **800** is in the off-state is that the contrast (the difference between on-state

lighting versus off-state lighting of the MEMS device **800**) is reduced. With the offset mirror **811**, the amount of second outgoing light **942B** directed to the projection aperture **948** is reduced compared to when a non-offset mirror is used. This is because the offset mirror **811** reduces exposure of a reflective target component (e.g., the cantilever hinge **808**, the spring tips **816A** to **816C**, the first raised electrode **812A**, and/or the second raised electrode **812B**) to the projection aperture **948**. In the example of FIG. **9B**, the second outgoing light **942B** is represented as being reflected from the spring tip **816A**. In other examples, outgoing light directed to the projection aperture **448** is due to stray light being reflected from another spring tip (e.g., the spring tip **816B**, the spring tip **816C**), a raised electrode (e.g., the first raised electrode **812A** or the second raised electrode **812B**), and/or the cantilever hinge **808**.

[0070] In the example of FIG. **9B**, the offset mirror **811** is offset in the $-X3$ direction to cover more of a target mechanical layer component (e.g., the spring tips **816A** to **816C**, first raised electrode **812A**, the second raised electrode **812B**, or the cantilever hinge **808** as in FIG. **8**), a target electrode layer component (e.g., the first address electrode **802A**, the second address electrode **802B**, and/or the bias electrode **804**), and/or a via (e.g., the mirror via **818**, the hinge vias **806**, the spring tip vias **314A**, **314B**, and **314C**, first electrode via **810A** and/or the second electrode via **810B**).

[0071] FIG. **10** is a top view **1000** of the offset mirror **811** of the MEMS device **800** of FIG. **3B** relative to the center position **809**. In the example of FIG. **10**, the center position **809** has the center point **813** and sides A, B, C, and D. The offset mirror **811** also has a center point **1012** and sides A, B, C, and D. In some examples, the center position **809** aligns the center point **813** with a mirror via center (e.g., a center point of the mirror via **818** in FIG. **8**) and/or a layer center (e.g., a center point of the mechanical layer of the MEMS device **800** in FIG. **8** and/or a center point of the electrode layer of the MEMS device **800** in FIG. **8**). As another option, the center position **809** is aligned with boundaries **1030A** and **1030B** in FIG. **10**. The boundaries **1030A** and **1030B** are selected, for example, based on outer boundaries of the electrode layer of the MEMS device **800** in FIG. **8** and/or of the mechanical layer of the MEMS device **800** in FIG. **8**. In the example of FIG. **10**, the offset mirror **811** is offset from the center position **809** or the center point **813** of the center position **809** in the $-X3$ direction by an offset (OFFSET2). OFFSET2 is also the difference between the center point **813** of the center position **809** and the center point **1012** of the offset mirror **811**. With OFFSET2, side A of center position **809** is offset from side A of the offset mirror **811** by $a2$, side B of center position **809** is offset from side B of the offset mirror **811** by $a2$, side C of center position **809** is offset from side C of the offset mirror **811** by $a2$, and side D of center position **809** is offset from side D of the offset mirror **811** by $a2$. Assuming a square shape for the offset mirror **811** and a perpendicular intersection, the distance from the center point **813** of the center position **809** to side A of the offset mirror **311** is $L/2 - a2$, where L is the length of each side of the offset mirror **811**. The distance from the center point **813** of the center position **809** to a perpendicular intersection of side B of the offset mirror **311** is $L/2 + a2$. The distance from the center point **813** of the center position **809** to a perpendicular intersection of side C of the offset mirror **811** is also $L/2 + a2$. The distance from the center point **813** of the center position **809** to a perpendicular intersection of side D of the offset mirror **811** is $L/2 - a2$. In the example of FIG. **10**, $\text{OFFSET2} = \sqrt{2} * a2$.

[0072] With OFFSET2 in the $-X3$ direction and related spacings $a2$, the offset mirror **811** improves coverage of a target mechanical layer component (e.g., the first raised electrode **812A**, the second raised electrode **812B**, the cantilever hinge **808**, and/or springs tips **816A** to **816C**), a target electrode layer component (e.g., the first address electrode **802A**, the second address electrode **802B**, and/or the bias electrode **804**), and/or a via (e.g., the mirror via **818**, the hinge vias **806**, the spring tip vias **314A** to **314C**, the first electrode vias **810A** and/or the second electrode vias **810B**) relative to a projection aperture. In some examples, OFFSET2 in the $-X3$ direction is approximately 0.318 micrometers and the spacing $a2$ is approximately 0.225 micrometers for a 5.4 μm pixel size. In this first example, OFFSET2 is approximately 5.89% of the pixel size. In another

examples, OFFSET2 in the -X3 direction is approximately 0.389 micrometers and the spacing a2 is approximately 0.275 micrometers for a 5.4 μm pixel size. In this second example, OFFSET2 is approximately 7.2% of the pixel size. In different examples, the offset, the spacing a1, and/or the pixel size may vary. Example pixel sizes may range from 16 μm pixels down to 2.7 μm pixels or smaller. FIGS. 11A to 11D are different views 1100, 1110, 1120, and 1130 of a pixel array 1102 based on MEMS device 800 with respective offset mirrors 811A to 811E. Each of the MEMS devices 800A to 800E is an example of the MEMS device 800 in FIG. 8. Each of the respective offset mirrors 811A to 811E is an example of the offset mirror 811 in FIG. 8. In FIGS. 11A to 11D, the offset mirrors 811A to 811E are offset in the -X3 direction relative to a center position (e.g., the center position 809 in FIGS. 8 and 10).

[0073] FIG. 11A is an incoming light angle view 1100 of part of the pixel array 1102 with the offset mirrors 811A to 811E in an on position (the mirror land direction or tilt for the on position is in the down direction in FIG. 11A). In the example of FIG. 11A, the offset mirrors 811A to 811E, and electrode layer component portions 1104A to 1104E of the respective MEMS devices 800A to 800E are represented. In FIGS. 11A to 11D, the offset mirrors 811A to 811E are offset relative to other MEMS device layers including respective electrode layers (e.g., each respective electrode layer of a MEMS device such as the MEMS device 800 in FIG. 8) and respective mechanical layers (e.g., each respective mechanical layer of a MEMS device such as the MEMS device 800 in FIG. 8). Each of the electrode layer component portions 1104A to 1104E may be part of a first address electrode (e.g., the first address electrode 802A in FIG. 8), part of a second address electrode (e.g., the second address electrode 802B in FIG. 8), or part of a bias electrode (e.g., the bias electrode 804 in FIG. 8).

[0074] FIG. 11B is a projection aperture view 1100 of the pixel array 1102 with the offset mirrors 811A to 811E in an on position (the mirror land direction or tilt for the on position is in the down direction in FIG. 11B). In the example of FIG. 11B, the offset mirrors 811A to 811E, the electrode layer component portions 1104A to 1104E, and mechanical layer component portions 1108A to 1108E of the respective MEMS devices 800A to 800E are represented. Each of the mechanical layer component portions 1108A to 1108E may be part of a cantilever hinge (e.g., the cantilever hinge 808 in FIG. 8), part of a spring tip (e.g., the spring tips 816A to 816C in FIG. 8), part of a first raised electrode (e.g., the first raised electrode 812A in FIG. 8), or part of a second raised electrode (e.g., the second raised electrode 812B in FIG. 8).

[0075] FIG. 11C is an incoming light angle view 1120 of the pixel array 1102 with the offset mirrors 811A to 811E in an off position (the mirror land direction or tilt for the off position is in the left direction in FIG. 11C). In the example of FIG. 11C, the offset mirrors 811A to 811E, the electrode layer component portions 1104A to 1104E, and the mechanical layer component portions 1108A to 1108E of the respective MEMS devices 800A to 800E are represented.

[0076] FIG. 11D is a projection aperture view 1130 of the pixel array 1102 with the offset mirrors 811A to 811E in the off position (the mirror land direction or tilt for the off position is in the left direction in FIG. 11D). In the example of FIG. 11C, the offset mirrors 811A to 811E, the electrode layer component portions 1104A to 1104E, and the mechanical layer component portions 1108A to 1108E of the respective MEMS devices 800A to 800E are represented.

[0077] In the incoming light angle view 1100 of FIG. 11A, with the offset mirrors 811A to 811E in the on position, most MEMS device components below the offset mirrors 811A to 811E are not visible except some electrode layer component portions such as the electrode layer component portions 1104A to 1104E. In the projection aperture view 1110 of FIG. 11B, with the offset mirrors 811A to 811E in the on position, most MEMS device components below the offset mirrors 811A to 811E are not visible except some electrode layer component portions such as the electrode layer component portions 1104A to 1104E and some mechanical layer component portions such as the mechanical layer component portions 1108A to 1108E. In the incoming light angle view 1120 of FIG. 11C, with the offset mirrors 811A to 811E in the off position, most MEMS device components

below the offset mirrors **811A** to **811E** are not visible except some electrode layer component portions such as the electrode layer component portions **1104A** to **1104E** and some mechanical layer component portions such as the mechanical layer component portions **1108A** to **1108E**. Although more of the electrode layer component portions **1104A** to **1104E** and more of the mechanical layer component portions **1108A** to **1108E** are visible in the incoming light angle view **1120** of FIG. **11C**, the incoming light view **1120** does not significantly affect the amount of stray light directed to a projection aperture. Exposure of electrode layer component portions **1104A** to **1104E** and/or the mechanical layer component portions **1108A** to **1108E** relative to a projection aperture is more relevant than exposure from an incoming light view. In the projection aperture view **1130** of FIG. **11D**, with the offset mirrors **811A** to **811E** in the off position, most MEMS device components below the offset mirrors **811A** to **811E** are not visible except some electrode layer component portions such as the electrode layer component portions **1104A** to **1104E** and some mechanical layer component portions such as the mechanical layer component portions **1108A** to **1108E**. Compared to mirrors aligned with center positions (e.g., the center position **809** in FIGS. **8** and **10**), the offset mirrors **811A** to **811E** cover more electrode layer component portions (e.g., more of the electrode layer component portions **1104A** to **1104E**) and/or mechanical layer component portions (e.g., more of the mechanical layer component portions **1108A** to **1108E**) relative to a projection aperture, which improves contrast.

[0078] Compared to mirrors in a centered position, the offset mirrors **811A** to **811E** in FIGS. **11A** to **11D** cover more mechanical layer component portions such as the mechanical layer component portions **1108A** to **1108E**. Compared to mirrors aligned with a center position, the offset mirrors **811A** to **811E** in FIGS. **11A** to **11D** may also cover more electrode layer component portions such as the electrode layer component portions **1104A** to **1104E**. To improve contrast of a display or projector, increased coverage of mechanical layer components for the projection aperture view **1130** of FIG. **11D** has highest priority. This is because scattered light due to visibility of mechanical layer components in the projection aperture view **1130** reaches the aperture (e.g., the projection aperture **138**). While full coverage of mechanical layer components for the projection aperture view **1130** is optimal, any coverage improvement of mechanical layer components (relative to centered mirrors) can provide a contrast improvement.

[0079] In some examples, electrode layer components are treated with a non-reflective coating, while mechanical layer components (e.g., the cantilever hinge **808**, the spring tips **816A** to **816C**, the first raised electrode **812A**, and/or the second raised electrode **812B**) are untreated. In some examples, treating mechanical layer components with a non-reflective coating is avoided to ensure consistent movement of the mechanical layer components. In the described examples, the reflectivity of untreated mechanical layer components may result in some extraneous reflected light being directed to a projection aperture. In the described examples, the amount of extraneous reflected light due to mechanical layer components is reduced, at least in a projection aperture view such as the projection aperture view **1130**, using offset mirrors (e.g., the offset mirrors **811A** to **811E**).

[0080] FIG. **12** is a top view **1200** of sculpted offset mirrors **1211A**, **1211B**, and **1211C** in accordance with various examples. In some examples, pixels and related mirrors are organized into reset groups. In such examples, sculpted offset mirrors, such as the sculpted offset mirrors **1211A**, **1211B**, and **1211C**, are sculpted in the direction of and the location of the reset groups. Each of the sculpted offset mirrors **1211A**, **1211B**, and **1211C** is an example of an offset mirror of the pixel array **136** in FIG. **1**, the offset mirror **211** in FIGS. **2A** and **2B**, the offset mirror **311** in FIGS. **3A**, **3B**, **4**, **5**, and **6**, the offset mirrors **311A** to **311E** in FIGS. **7A** to **7E**, the offset mirror **811** in FIGS. **8**, **9A**, **9B**, and **10**, or any of the offset mirrors **811A** to **811E** in FIGS. **11A** to **11D**. In the example of FIG. **12**, the sculpted offset mirror **1211B** includes a sculpted top edge **1208** and a sculpted bottom edge **1210**. In other examples, all four edges of sculpted offset mirror **1211B** may be sculpted. The sculpted offset mirror **1211A** has a sculpted bottom edge **1212**, and the sculpted offset mirror

1211C has a sculpted top edge **1214**. The sculpted edges have a concave shape in this example. Because of the sculpted top edge **1208** and the sculpted bottom edge **1212**, the gap between sculpted offset mirrors **1211A** and **1211B** is larger in the center along the sculpted top edge **1208** and the sculpted bottom edge **1212** than at the ends of the sculpted top edge **1208** and the sculpted bottom edge **1212**. The middle gap **1216** is larger than the corner gaps **1218A** and **1218B**. The middle gap **1216** is a first gap between a first point on sculpted top edge **1208** and the nearest point to the first point on the sculpted offset mirror **1211B**. The middle gap **1216** is larger than the corner gap **1218A** between a second point on the sculpted top edge **1208** and a nearest point to the second point on the sculpted offset mirror **1211B**.

[0081] In some examples, the sculpted offset mirrors **1211A**, **1211B**, and **1211C** have a larger gap in the middle so an appropriate distance can be maintained between the edges of the sculpted offset mirrors **1211A**, **1211B**, and **1211C** to prevent damage during operation. In FIG. **12**, corner gaps **1218A** and **1218B** are smaller than the middle gap **1216**. Smaller corner gaps allow for a larger total surface area of an SLM to be covered by offset mirrors. As described above, gaps allow light to shine between the offset mirrors, which can cause optical scatter and also heat up the understructure of a MEMS device, which affects the overall temperature of the device. The sculpted top edge **1208** and the sculpted bottom edge **1212** of the sculpted offset mirrors **1211A** and **1211B**, respectively, provide the larger gap (e.g., the middle gap) where it is most useful while also providing a smaller gap at the corners to cover a larger total surface area of an SLM with offset mirrors. The ratio between the “useful” (reflective) area of a SLM and the complete area of the array, which includes the gaps and the vias, is known as the on-state fill factor. The fill factor is the fraction of a SLM pixel that is actually reflective. A sculpted offset mirror array helps to protect against damage caused by gaps that are too small while also providing a higher fill factor than pixel arrays that have gaps that are too large.

[0082] In the example of FIG. **12**, the sculpted offset mirror **1211B** also has a sculpted bottom edge **1210** that is adjacent to the sculpted top edge **1214** of the sculpted offset mirror **1211C**. Because of these sculpted edges, the middle gap **1220** is larger than corner gaps **1222A** and **1222B**. In some examples, the middle gap **1216** (e.g., a first gap) between a first center of a first edge (the sculpted top edge **1208**) and the sculpted offset mirror **1211A**. The middle gap **1216** is larger than a corner gap **1222A** (e.g., a second gap) between a first end of the sculpted top edge **1208** and the sculpted offset mirror **1211A**.

[0083] FIG. **13** is a flowchart showing a MEMS device fabrication method **1300** in accordance with various examples. The MEMS device fabrication method **1300** includes forming an electrode layer at block **1302**. At block **1304**, a mechanical layer is formed over the electrode layer, the mechanical layer including a reflective material. At block **1306**, a mirror is formed over the mechanical layer, the mirror is offset relative to the mechanical layer to reduce reflections from the reflective material directed to a projection aperture (e.g., the projection aperture **138** in FIG. **1**). With the offset mirror formed at block **1306**, the amount of reflections from the reflective material that are directed towards the projection aperture is reduced as described herein.

[0084] In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

[0085] Also, in this description, the recitation “based on” means “based at least in part on.” Therefore, if X is based on Y, then X may be a function of Y and any number of other factors.

[0086] A device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the

function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

[0087] A circuit or device that is described herein as including certain components may instead be adapted to be coupled to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors, and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be adapted to be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture, for example, by an end-user and/or a third-party.

[0088] Circuits described herein are reconfigurable to include additional or different components to provide functionality at least partially similar to functionality available prior to the component replacement. Components shown as resistors, unless otherwise stated, are generally representative of any one or more elements coupled in series and/or parallel to provide an amount of impedance represented by the resistor shown. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in parallel between the same nodes. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in series between the same two nodes as the single resistor or capacitor.

[0089] While certain elements of the described examples are included in an integrated circuit and other elements are external to the integrated circuit, in other examples, additional or fewer features may be incorporated into the integrated circuit. In addition, some or all of the features illustrated as being external to the integrated circuit may be included in the integrated circuit and/or some features illustrated as being internal to the integrated circuit may be incorporated outside of the integrated circuit. As used herein, the term “integrated circuit” means one or more circuits that are: (i) incorporated in/over a semiconductor substrate; (ii) incorporated in a single semiconductor package; (iii) incorporated into the same module; and/or (iv) incorporated in/on the same printed circuit board.

[0090] In this description, unless otherwise stated, “about,” “approximately” or “substantially” preceding a parameter means being within ± 10 percent of that parameter or, if the parameter is zero, a reasonable range of values around zero.

[0091] Modifications are possible in the described examples, and other examples are possible, within the scope of the claims.

Claims

1. A microelectromechanical system (MEMS) device comprising: a mechanical layer; a mirror; and a mirror via coupling the mechanical layer and the mirror, wherein the mirror is laterally offset from the mechanical layer in a direction.
2. The MEMS device of claim 1, wherein the mechanical layer includes a torsion hinge, the mirror has an on position and an off position, and the mirror in the off position increases coverage of the torsion hinge relative to a projection aperture compared to a mirror aligned with a center position.
3. The MEMS device of claim 2, wherein the mirror is laterally offset along the direction by at least $0.15\text{ }\mu\text{m}$ relative to the center position.
4. The MEMS device of claim 1, wherein the mechanical layer includes a cantilever hinge, the mirror has an on position and an off position, and the mirror in the off position increases coverage of the cantilever hinge relative to a projection aperture compared to a mirror aligned with a center

position.

5. The MEMS device of claim 4, wherein the mirror is laterally offset along the direction by at least $0.225\text{ }\mu\text{m}$ relative to the center position.

6. The MEMS device of claim 1, wherein the mechanical layer includes a spring tip, the mirror has an on position and an off position, and the mirror in the off position increases coverage of the spring tip relative to a projection aperture compared to a mirror aligned with a center position.

7. The MEMS device of claim 1, wherein the mirror via is centered relative to the mechanical layer.

8. The MEMS device of claim 1, wherein the mirror has sculpted edges.

9. A microelectromechanical system (MEMS) device comprising: an electrode layer; a mechanical layer; hinge vias coupling the electrode layer and the mechanical layer; a mirror; and a mirror via coupling the mechanical layer and the mirror, the mirror having a first position when the MEMS device is in an on state, the mirror having a second position when the MEMS device is in an off state, and the mirror laterally offset relative to the mechanical layer in a direction.

10. The MEMS device of claim 9, wherein the mechanical layer includes a torsion hinge, the mirror has an on position and an off position, and the mirror in the off position increases coverage of the torsion hinge relative to a projection aperture compared to a mirror aligned with a center position.

11. The MEMS device of claim 9, wherein the mechanical layer includes a cantilever hinge, the mirror has an on position and an off position, and the mirror in the off position increases coverage of the cantilever hinge relative to a projection aperture compared to a mirror aligned with a center position.

12. The MEMS device of claim 9, wherein the mechanical layer includes a raised electrode, the mirror has an on position and an off position, and the mirror in the off position increases coverage of the raised electrode relative to a projection aperture compared to a mirror aligned with a center position.

13. The MEMS device of claim 9, wherein the mechanical layer includes a spring tip, the mirror has an on position and an off position, and the mirror in the off position increases coverage of the spring tip relative to a projection aperture compared to a mirror aligned with a center position.

14. The MEMS device of claim 9, wherein the mirror via is centered relative to the mechanical layer.

15. A microelectromechanical system (MEMS) device comprising: an electrode layer; a mechanical layer; hinge vias coupling the electrode layer and the mechanical layer; a mirror; and a mirror via coupling the mechanical layer and the mirror, wherein the mirror is laterally offset relative to the mechanical layer in a direction.

16. The MEMS device of claim 15, wherein the mechanical layer includes a torsion hinge and spring tips, the mirror has an on position and an off position, and the mirror in the off position increases coverage of the torsion hinge relative to a projection aperture compared to a mirror aligned with a center position.

17. The MEMS device of claim 15, wherein the mechanical layer includes a cantilever hinge and a spring tip, the mirror has an on position and an off position, and the mirror in the off position increases coverage of the cantilever hinge relative to a projection aperture compared to a mirror aligned with a center position.

18. The MEMS device of claim 15, wherein the mechanical layer includes a raised electrode, the mirror has an on position and an off position, and the mirror in the off position increases coverage of the raised electrode relative to a projection aperture compared to a mirror aligned with a center position.

19. The MEMS device of claim 15, wherein the mechanical layer includes a spring tip, the mirror has an on position and an off position, and the mirror in the off position increases coverage of the spring tip relative to a projection aperture compared to a mirror aligned with a center position.

20. The MEMS device of claim 15, wherein the mirror via is centered relative to the mechanical layer.
